

Diffusion-MoE Phase 2 Training Investigation

Findings report — all-layers training run, bugs found, and current state

Prepared for review · diffusion-moe project

Executive Summary

Following Phase 1's diagnostics (`phase1_findings_report.md`) and the bounded single-layer pilot fine-tune, this phase asked the question those tools structurally couldn't answer: **what actually happens if every layer of the project's own from-scratch architecture is a `DiffusionMoELayer` , trained end-to-end, not just diagnosed or spliced into a frozen pretrained backbone?**

The honest answer is: six real, previously-undiscovered bugs, one badly-miscalibrated hyperparameter, and — after fixing all of them — clean, complete training runs that demonstrate the architecture can learn, route without permanently collapsing, and finish without crashing. That is real progress, not a finished result: a same-budget dense-model control, repeated across two seeds, decisively and consistently outperformed the all-layers diffusion-MoE configuration on quality, cost, *and* reliability (§6.4–6.5). See §7 for the full picture and what would change that conclusion.

1. **NaN divergence at step ~199** — `NystromDiffusionMap` 's kernel bandwidth (`eps_`) is fit once and frozen between landmark refits; once training moves the embedding distribution far enough, a token can drift outside every landmark's kernel support, and the unguarded division silently produces NaN, which `clip_grad_norm_` cannot repair. Fixed with both a defensive guard and — the deeper fix — a live `eps_` refresh between refits (§2.1).
2. **Checkpoint resume crashes a real MoE layer** — `NystromDiffusionMap` 's fitted state lives on a plain Python object, not an `nn.Module` , so it silently does not survive `state_dict()` save/load. Any resume off the exact refit-schedule boundary crashed outright. Found running the project's own `expert_attribution.py` tool against a real checkpoint, not by inspection (§2.2).
3. **No shared/always-active expert** — every FFN computation was contingent on sparse top-k routing; nothing had guaranteed, routing-independent gradient signal. Added a mandatory `shared_expert` , architecturally identical to a routed `ExpertFFN` , applied unconditionally outside the router (§3).
4. **Disk-full crash near the finish line** — unbounded checkpoint retention filled a 60GB instance's disk with ~5.7GB files, corrupting the final checkpoint write. Fixed with automatic pruning (§4).
5. **A catastrophic initialization bug** — `task_loss` started at ~823 nats against a random-guessing baseline of $\ln(32000)=10.37$, and never recovered below ~60 nats across any run, no matter how long training continued. Root-caused to two compounding, previously-unaddressed initialization gaps (no custom init existed anywhere in the model) and fixed; confirmed both in isolation and in a real training run (§5). This is the most consequential finding in this phase — everything observed in earlier runs (§6) was training on top of this bug.
6. **`mu_load` (the load-balance loss weight) was miscalibrated by ~2 orders of magnitude** relative to `task_loss` 's gradient once the loss scale was actually correct — measured directly, not guessed, and retuned (§6.2).
7. **`cfg.seed` was a no-op** — never applied to torch's global RNG anywhere in this pipeline (only `data.seed` , controlling shuffle order, was wired through), so model weight initialization — the dominant source of run-to-run variation — drew from whatever unseeded state the process happened to start in. Fixed, and what made the seed-repeat check in §6.5 meaningful in the first place (§9).

The final MoE run (`graceful-universe-21`) completed its full 2,441-step / 20M-token budget cleanly: zero NaN, `task_loss` ending at 7.08 (below the random baseline), all 24 layers' `load_loss` ending near zero (down from 7 of 24 near-total collapse mid-run), and `val_pp1` finally moving off its `exp(20)` safety cap for the first time in this investigation (7,286 → 5,081 → 35,294 spike → 16,158 → 3,245 → 2,239 → 974). That is real progress relative to every earlier run in this phase — but a same-budget **dense-model control run** (`peachy-cloud-22` , \$6.4), run immediately afterward, decisively beat it: `task_loss=4.80` , `val_pp1=93.2` , in ~15x less wall-clock time. A second seed of both configs (§6.5) confirmed this wasn't one lucky/unlucky comparison — dense stayed essentially unchanged (sub-1% variance), while the MoE run got worse on the second seed (`val_pp1=1,478` , wall-clock 425 min) rather than better. Read together, this phase's honest bottom line is: the infrastructure now works, and the current all-layers diffusion-MoE configuration is decisively and consistently worse than the simple baseline it needs to beat. See §7.

1. Background and Motivation

Phase 1's diagnostics and the bounded pilot fine-tune (`phase1_findings_report.md` §3.4–3.6) each tested one layer at a time, spliced into an otherwise-frozen pretrained Mistral-7B backbone. That structurally cannot answer a sharper question raised during review: *if a layer's router keeps forcing collapse onto one expert during real training, is that proof the layer can't benefit from diffusion routing at all, or just proof that one frozen-backbone, one-layer splice hasn't given the router anything to differentiate?* The only way to find out is to train the project's own from-scratch architecture with every layer routed, and watch. The user's explicit direction: *"I would let all layers use diffusion MoE and see what happens."*

This phase covers that run, the infrastructure it required (per-layer loss breakdown, checkpoint resume, expert-attribution analytics — see `methodology.md` §5 and §6 for the durable reference material), the bugs it surfaced, and the final result.

All runs in this section use the project's own 300M/24-layer from-scratch architecture (`configs/model/300m.yaml`), trained on `wikitext-103-raw-v1` via local parquet files (bypassing the recurring flaky-CDN Hub-streaming stall — see `methodology.md` §4), on a single rented RTX 3090 instance (vast.ai).

2. Bug: NaN Divergence, and the Deeper Fix Behind It

2.1 Root cause: a frozen kernel bandwidth under a moving distribution

The first all-layers run diverged to NaN at step 199, confirmed via an explicit `RuntimeWarning: invalid value encountered in divide` in `geometry/nystrom.py` . `NystromDiffusionMap.transform()` 's Nyström extension formula divides by each new point's total kernel mass to the (frozen) landmarks:

```
K_alpha = K_new / (d_alpha_new[:, None] * self._d_alpha_landmarks[None, :])
```

`eps_` (the kernel bandwidth) is derived once from the landmarks' own spread at the last refit (`fit()`), only run every `centroid_refresh_steps`) and then frozen until the next one. If a token's embedding drifts far enough from that frozen reference — plausible once `centroid_separation_loss` has been actively pushing centroids apart for hundreds of steps — its kernel value to every landmark can underflow to exactly `0.0` in float64 (`exp(v)` underflows to the literal bit pattern `0.0` below roughly `v < -745` , not a small-but-nonzero value). The resulting `0/0` division produced NaN , which propagated through `task_loss` / `load_loss` , and — critically — `torch.nn.utils.clip_grad_norm_` cannot repair a NaN gradient (the norm of a NaN tensor is itself NaN), so every parameter update after that point silently corrupted the entire model. Training continued to "run" for ~40 more steps on fully poisoned weights before

crashing outright on an unrelated downstream `KMeans` call (sklearn's input validator rejects NaN explicitly; the raw numpy division earlier did not).

Immediate fix: guard the division (`d_alpha_new_safe` , mirroring the existing `row_sums` guard a few lines below) so a token with zero kernel mass gets diffusion coordinates at the origin (maximal uncertainty) instead of NaN.

2.2 The real fix: live-refreshing the kernel bandwidth between refits

The guard alone treats a symptom. The actual staleness problem — `eps_` describing a step-0 snapshot of the embedding distribution, growing more wrong the longer training runs between refits — remains. Fixed by having `transform()` re-derive `eps_` from the *current* batch's actual distance to the (still-frozen) landmarks on every call, reusing the distance matrix already computed for the Nyström kernel (cheap — no new k-means).

This surfaced two further correctness requirements, both needed for the fix to be mathematically sound rather than merely avoid a crash:

- **eigenvectors_ / psi_landmarks_ must be re-derived alongside `eps_`** (extracted into a shared `_fit_landmark_spectrum` , called by both `fit()` and the refresh path): Nyström's defining correctness property — that applying the extension formula to the landmarks themselves must exactly reproduce their direct diffusion coordinates — only holds when the landmark spectrum and the new-point extension share one consistent kernel. Caught by `test_nystrom_quality_exact_reproduction_on_landmarks` when a first attempt (refreshing only `eps_` in isolation) broke it (0.0042 error against a $1e-4$ tolerance).
- **diffusion_eigenvectors 's ARPACK solve needed a seeded starting vector.** It previously ran once per landmark set, so its default unseeded random start never mattered. Once it could rerun every step, an unseeded start would make the diffusion-coordinate basis jitter step-to-step from solver randomness alone — independent of any real drift in the data, i.e. the opposite of what this fix was for.
- **fit_transform 's own internal `transform()` call must skip the refresh** (new `_refresh_eps` parameter): it runs immediately after `fit()` on the same data, so there is no staleness yet to correct, and refreshing anyway would silently override `fit()` 's landmark-based heuristic and redo the eigensolve twice on every refit step for no benefit.

Cost: this is not free, but it is cheap relative to what it replaces. `eps_` 's refresh reuses an already-computed distance matrix; the re-derived eigensolve is over just the `n_landmarks × n_landmarks` matrix (32–128 points), not another k-means clustering over the full token batch. Measured directly in production: steady-state per-step time rose from ~5.55s to ~7.0–7.3s (roughly 25–30%), on top of an already CPU-bound geometry pipeline (this entire subsystem runs on `z.detach()` 'd numpy/sklearn, never touching the GPU).

3. Feature: A Mandatory Shared Expert, Outside the Router

Once every layer was routed and training could be observed cleanly, a striking pattern emerged (visible even before the initialization bug was found, §5): `load_loss` stayed elevated across nearly all 24 layers simultaneously — not one collapsed outlier, a near-universal, if partial, imbalance (mean `load_loss` 3.21 across all layers at one snapshot, only 1 of 24 layers below 1.0). This raised the mechanistic question motivating this whole run: with every FFN computation contingent on sparse top-k routing, nothing in the network had guaranteed, routing-independent gradient signal — common/generic computation had to be either redundantly relearned by whichever expert won that step's routing lottery, or not learned reliably at all if routing was noisy.

This is a known pattern in production MoE architectures (DeepSeekMoE's "shared expert isolation," similarly Qwen2-MoE), added here as `DiffusionMoELayer.shared_expert` : architecturally identical to a routed `ExpertFFN` (same width, same `overlap_factor`), applied unconditionally to every token, every step — no gate value, no centroid, entirely outside the diffusion router's influence. This adds real active compute per token (`top_k + 1` experts fire instead of `top_k`), not a reallocation of existing capacity.

Verified with a direct behavioral test

(`test_shared_expert_contributes_even_when_routing_contributes_nothing`): with the routed dispatch path monkeypatched to contribute exactly zero, the layer's output still differs from the plain attention residual — the shared expert's contribution cannot be made routing-contingent by construction.

4. Bug: Disk-Full Crash Near the Finish Line

A run that had otherwise reached step 2,400 of 2,441 (98.3%) crashed on its own checkpoint save: the instance's 60GB disk filled completely (8 accumulated checkpoints × ~5.7GB each), and the final write was cut off mid-file (`PytorchStreamWriter failed writing file` ; the resulting `step_2400.pt` was truncated to ~2.97GB against an expected ~5.68GB — corrupt, unusable). The last *valid* checkpoint (`step_2100.pt` , saved before the disk filled) was confirmed to load cleanly.

Fixed with automatic checkpoint retention in `Trainer` : after each numbered checkpoint save, all but the most recent `keep_last_n_checkpoints` (default 2, configurable via `training.keep_last_n_checkpoints`) are deleted. `best.pt` is a separate file and always kept regardless. `<= 0` disables pruning (old, unbounded behavior) for anyone who wants it.

5. The Initialization Bug: Task_Loss Was Never Actually Calibrated

5.1 The anomaly

Every run before this fix showed the same signature: `task_loss` starting around 823 nats (step 1) and never recovering below roughly 60 nats, however long training continued. `val_ppl ( exp(min(avg_nll, 20.0)) )` was pinned at its safety cap — 485,165,195.4 exactly — at *every single logged evaluation, across every run, all the way to the end of a 2,400-step run*. This is a real red flag independent of routing behavior: this project's tokenizer (Mistral's, 32,000 tokens) gives a random-guessing baseline of `ln(32000) ≈ 10.37` nats — the cross-entropy a properly-initialized, completely untrained model should sit near. A trained model that never gets within even a factor of 2 of that baseline, in either direction, is not "undertrained" — something is numerically broken.

5.2 Root cause, confirmed by direct measurement

No custom weight initialization existed anywhere in the model — every linear layer and the token embedding relied entirely on PyTorch's defaults. Two compounding gaps, confirmed empirically (not just inferred) by hooking every block of a freshly-constructed 24-layer model and measuring residual-stream norms and final logit magnitude directly:

1. `token_embedding` used `nn.Embedding` 's default init — `Normal(0, 1)` , `std=1.0` — far too large for a transformer embedding table, and especially costly because `tie_embeddings=True` by default makes this same oversized matrix double as the LM-head unembedding projection, directly inflating logit magnitude. This was the *dominant* cause: fixing only the second gap below (residual-output scaling) left final logits unchanged — actually slightly worse (`max abs` 828 → 1071) — until this was also fixed.

2. **No depth-aware scaling on residual-branch output projections** (attention's `out_proj`, every FFN's `down_proj` — dense `TransformerBlock`, routed `ExpertFFN`, and the new `shared_expert` alike). Every serious transformer implementation (GPT-2, LLaMA, nanoGPT) scales these down by $1/\sqrt{2 \times n_layers}$ at init specifically because residual-stream variance otherwise compounds with depth. Confirmed directly: residual-stream norm grew $32 \rightarrow 44$ over 24 layers unfixed; a controlled, smooth $1.0 \rightarrow 7.1$ fixed.

Combined fix: `token_embedding` initialized to the standard `std=0.02` (GPT-2/nanoGPT convention); every `out_proj` / `down_proj` weight scaled by $1/\sqrt{2 \times n_layers}$ after construction (matched on the Linear's own attribute name, so it applies uniformly regardless of which module — dense, routed expert, or shared expert — it's nested in).

Verified twice. In isolation: a fresh random-init model's cross-entropy against random labels dropped from $\sim 1,014$ to **10.51** (essentially exactly $\ln(32000)=10.37$). In real training (`vital-microwave-20`, the first run with this fix): step 1 `task_loss` = **10.61**, and a clean, sane trajectory afterward ($10.61 \rightarrow 9.03 \rightarrow 8.47$ by step 26) — nothing resembling any earlier run's trajectory.

5.3 What this means for everything observed before this fix

`resilient-brook-18` and `hardy-lake-19` (§6) both trained on top of this bug. Their qualitative conclusions (per-layer collapse patterns, shared-expert comparison) are likely still valid as *relative* signals within those runs, but their absolute loss numbers should not be treated as meaningful, and the shared-expert-vs-not comparison in §6.2 should be re-run now that the loss scale is correct before drawing any firm conclusion from it.

6. Run-by-Run Summary

Run (wandb)	Init fix	Shared expert	<code>mu_load</code>	Outcome
<code>resilient-brook-18</code>	No	No	0.01	<code>task_loss</code> plateaued 60–110 nats; <code>val_pp1</code> permanently capped; killed to add the shared expert
<code>hardy-lake-19</code>	No	Yes	0.01	Modest improvement over the above at matched steps (<code>task_loss</code> =90.3 vs 100.3 @ step 600) but still broken-scale; crashed on disk-full at step 2,400/2,441 (§4)
<code>vital-microwave-20</code>	Yes	Yes	0.01	Step 1 <code>task_loss</code> =10.61 (correct!), but 18 of 24 layers collapsed to <code>load_loss</code> ≈7.0 (the exact CV ² ceiling) by step ~600; <code>task_loss</code> climbed back to 11.5 — worse than its own starting point; killed for <code>mu_load</code> retuning
<code>graceful-universe-21</code>	Yes	Yes	2.0	Completed cleanly — see §6.3
<code>peachy-cloud-22</code> (dense control, <code>layers_to_replace=[]</code>)	Yes	n/a	n/a	Same budget, same init. Beat graceful-universe-21 decisively — see §6.4
<code>sunny-surf-23</code> (dense, <code>seed=123</code>)	Yes	n/a	n/a	Near-identical to <code>peachy-cloud-22</code> — see §6.5

cool-pyramid-24 (MoE, seed=123)	Yes	Yes	2.0	Completed cleanly, but meaningfully worse and slower than graceful-universe-21 — see §6.5
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6.1 The collapse dynamic got *worse*, not better, once the loss was fixed

`vital-microwave-20` is the sharpest result in this phase. Fixing the initialization did not just fix the loss scale — it let gradients flow cleanly for the first time. `task_loss` 's gradient became a strong, well-calibrated signal instead of one dominated by first fixing a catastrophic starting point, which made the router-collapse shortcut (route everything to one expert per layer, reducing the effective optimization problem) *more* attractive to find, not less. 18 of 24 layers reached essentially the exact `load_loss` ceiling ($n_experts - 1 = 7.0$) within ~600 steps. This is a direct escalation of the question this whole run was designed to answer: collapse is not a rare pathology in one unlucky layer, it is close to the *default* outcome once the network can actually learn efficiently, unless something resists it hard enough.

6.2 Measuring, not guessing, the fix

Rather than guess a new `mu_load` , the relative gradient pull of `task_loss` vs. `load_loss` on the shared `ExpertCentroids` parameter was measured directly on a fresh, correctly-initialized model (the parameter both losses actually compete to move, since `load_loss` 's gradient is confined to centroids — `Psi_t` is detached before the routing-loss computation, so it never reaches attention/backbone weights directly). Raw ratio: `task_loss`'s pull was only **1.86×** `load_loss`'s — a fair fight. The old `mu_load=0.01` shrank that to an effective **186×**, leaving `load_loss` almost no real ability to resist collapse — a value very likely tuned against the old, catastrophically-mis-scaled `task_loss` and never revisited once §5's fix corrected it. Set to `mu_load=2.0` , targeting an effective ratio near 1.

6.3 The final run

`graceful-universe-21` completed the full 2,441-step budget cleanly — zero NaN, `state=finished` . At the halfway point (step ~296), the picture was a real but partial improvement: 7 of 24 layers near collapse, versus 18 of 24 in the unfixed- `mu` run at a comparable point — down ~60%, not eliminated. By the *end* of training, the picture resolved further: all 24 layers' `load_loss` near zero (best 0.008, most below 0.0005) — though the path there was genuinely volatile, not a clean monotonic decrease (`load_loss` oscillated between 0.45 and 3.6 across steps 1,200–2,200 before settling as the learning rate decayed toward its floor; see §7 for why this specific ending should not yet be over-interpreted).

`task_loss` ended at **7.08** — below the random baseline, genuine learning. `val_ppl` moved for the first time in this entire investigation, off the `exp(20)` cap:

```
step 300: 7,286
step 600: 5,081
step 900: 35,294  <- real spike, not a fluke
step 1200: 16,158
step 1800: 3,245
step 2100: 2,239
step 2400: 975   <- best of the run
```

Non-monotonic and still high in absolute terms for a 300M-parameter model (even on a small budget), but a real, measured signal that the model is generalizing at all — something no earlier run in this entire investigation had shown.

6.4 The dense-model control, and a decisive negative result

Immediately after `graceful-universe-21` completed, the same config was rerun with `layers_to_replace=[]` (dense baseline, identical data/budget/token count/hyperparameters otherwise) — the missing control flagged in every version of §7 until now. Same starting point confirmed (`task_loss=10.59` at step 1 vs. the MoE run's `10.61` , so the initialization fix applies uniformly and this is a fair comparison):

	Dense (peachy-cloud-22)	Diffusion-MoE, all layers (graceful-universe-21)
Final task_loss (step 2441)	4.80	7.08
Final val_ppl	93.2	974.6
val_ppl trajectory	Smooth, monotonic: 274 → 200 → 156 → 131 → 100 → 95 → 93	Noisy, huge spike at step 900: 7,286 → 5,081 → 35,294 → 16,158 → 3,245 → 2,239 → 975
Wall-clock, same 2,441 steps	20.1 min	294.5 min (~4.9 hours, ~15x longer)

The dense model wins decisively on every axis, at matched token budget. The time gap is almost entirely the CPU-bound Nyström/eigensolve pipeline (~7 s/step MoE vs. ~0.49 s/step dense — matches the ~15x wall-clock ratio directly), not GPU compute. This is not even a "worse quality, but cheaper active compute" tradeoff in MoE's favor: the diffusion-MoE model's active FFN width per token (`top_k+1=3` experts × 512 each = 1,536) is *smaller* than the dense model's full 4,096-wide FFN — less active capacity, ~15x the wall-clock cost, and a meaningfully worse result.

This is a real, decisive answer, and it is unfavorable to the current approach. Nothing in this investigation shows the added complexity of diffusion-routed MoE buying anything at this scale/budget. The standard framing for why sparse MoE is worth its complexity (Clark et al. 2022's scaling-law work, referenced earlier in this investigation) is specifically about quality-per-active-FLOP *at scale* — 20M tokens on a 300M-parameter model is nowhere near the regime that literature has actually demonstrated an advantage in. This result is consistent with that: it does not show diffusion-MoE is a bad idea in general, but it does mean this specific comparison offers no evidence *for* it, and meaningful evidence against training it further at this scale without a specific, articulated reason to expect the gap to close only at larger scale.

6.5 Seed-repeat check: dense is consistent, diffusion-MoE is not

Both §6.4 configs were rerun with `seed=123` (a new fix this phase: `cfg.seed` was previously never actually applied to torch's global RNG — only `data.seed` , controlling shuffle order, was wired through, meaning every prior run's model initialization was drawn from whatever unseeded state the process happened to start in, not a controlled value at all — see §9).

	seed=42	seed=123	Spread
Dense task_loss	4.80	4.80	~0%
Dense val_ppl	93.2	92.8	<1%
Dense wall-clock	20.1 min	20.0 min	~0%

MoE task_loss	7.08	7.44	~5%
MoE val_ppl	974.6	1,477.5	~52%
MoE wall-clock	294.5 min	424.6 min	~44%
MoE load_loss (final)	0.00044	0.00020	both ≈0

Two things confirmed here. First, §6.4's dense-beats-MoE result is not a fluke of one lucky/unlucky seed pair — dense is close to seed-invariant, and the *worse* of the two MoE seeds (seed=123, val_ppl=1,478) is still over 15x worse than *either* dense run. Second, and a genuinely new finding rather than just confirmation: diffusion-MoE is not just worse on average, it is meaningfully *less consistent* than dense, on both quality and cost — a 52% perplexity swing and a 44% wall-clock-time swing between two runs that differ only in random seed, versus dense's sub-1% variation on both. This extends Phase 1 §3.5's seed-42-vs-123 finding (different experts collapsing depending on seed) rather than just replicating it: it's not only *which* expert collapses that's seed-sensitive, it's the *total training cost and final quality* too — a real, separate cost of the current approach's variance, independent of its average performance gap.

6.6 What token-level routing actually looks like (early-training snapshot)

`scripts/expert_attribution.py` (extended this phase with Hydra CLI override support, so it can point at whatever data config a checkpoint was actually trained against — see `methodology.md` §6) was run against `hardy-lake-19`'s step-300 checkpoint. Filtering out padding (a first pass mistakenly included it — 479 of 492 positions in the sampled sequence were padding, trivially uniform) revealed real, reproducible structure, but **mostly positional, not semantic**, at this early stage: layer 0 routes only the first 1–4 tokens of every sequence to one expert (sequence-start detection) and everything else to another, regardless of topic; layer 6 shows a clean depth-progression through three experts as a sequence gets longer, independent of content; layer 20 was fully collapsed (one expert, ~99% of all real tokens across 5 sampled sequences). One layer (7) showed a plausible content-correlated split — a block of numeric/statistical text routing differently than surrounding narrative prose — but with one sequence and no position control, this could not be distinguished from coincidence.

6.7 Which layers to select for a follow-up: a data-driven answer, this architecture's own

§7's revised recommendation points toward a *selective* layer-replacement strategy rather than all-layers. Phase 1's oracle-ceiling layer indices (10/29/30 recoverable; 17/20/31 needing cosine) don't transfer — that was a 32-layer Mistral-7B model, this is a 24-layer from-scratch one. This section derives candidates from this architecture's own data instead, combining both completed MoE runs (§6.5).

Per-layer `load_loss` , averaged across each run's full trajectory (not just the final point), both seeds:

Layer	seed=42 mean	seed=123 mean	Average	Seed-to-seed spread
0	0.223	0.206	0.214	0.017
1	0.901	1.328	1.114	0.427
2	0.769	1.701	1.235	0.932
4	1.792	1.890	1.841	0.098
17	2.344	2.159	2.252	0.185
21	3.909	3.743	3.826	0.167

20	3.328	3.601	3.464	0.273
22	4.182	3.194	3.688	0.988
23	4.264	3.307	3.786	0.957
3, 5, 9, 14, 19	—	—	mid-range	1.0–1.6 (most volatile)

(Remaining 13 layers omitted for space; the full 24-row table is reproducible from the wandb API, see §6.5's query.) Two independent signals matter here, not one: a low *average* means a layer tends to stay balanced; a low *spread* means that's reliable rather than a coin flip. Layer 0 dominates on both. Layers 4, 17, and 21 are a clear second tier — moderate-to-high averages, but consistently so (low spread) in both seeds. Layers 3/5/9/14/19 have seed-to-seed spreads of 1.0–1.6 — even where their average looks tolerable, they swing unpredictably and are poor candidates regardless of mean.

Confirmed against the final checkpoints directly, not just the training-average proxy —

`expert_attribution.py` 's top-expert-share on both runs' `step_2400.pt` :

Layer	seed=42 top share	seed=123 top share
0	63.9%	58.0%
4	78.6%	91.9% (collapsed)
17	91.4% (collapsed)	97.3% (collapsed)
20	64.2%	97.9% (collapsed)
21	69.2%	97.3% (collapsed)

Layer 0 is the only layer that stayed meaningfully balanced on *both* seeds at the final checkpoint — everything else that looked reasonable on seed=42 (4, 20, 21) collapsed hard on seed=123. This matches the training-average table's ranking exactly and is the strongest, most consistent signal in this investigation for which single layer to pick first.

What layer 0 actually learned, at the token level (padding filtered): on `graceful-universe-21` 's (seed=42) final checkpoint, layer 0 shows a clean, token-by-token split correlating with a real linguistic feature — numeric/digit tokens route to one expert, word tokens to another, consistently across all 5 sampled sequences (e.g. `2:5 3:5 @:5 .:5` vs. `the:1 city:1 population:1 was:1` in the same sequence). Layer 4 shows the same categorical split with the expert IDs swapped. The collapsed layers show either nothing (layer 17: ~everything to one expert, no structure at all) or a noisier, degraded version of the same digit/word signal (layers 20/21: mostly consistent, with real exceptions).

This does not fully reproduce on `cool-pyramid-24` 's (seed=123) final checkpoint, and the way it fails to reproduce is itself informative: layer 0 stays balanced (confirming the robustness finding above), but the *specific* boundary it draws is different — mostly one expert throughout, with a second expert taking over specifically in the numeric-dense passage plus scattered periods/conjunctions, rather than seed=42's clean per-token digit/word alternation. **The honest reading:** layer 0 reliably learns to *specialize rather than collapse* — that part is robust across seeds, confirmed three independent ways here (training-average `load_loss` , its spread, and final-checkpoint top-share). *What* it specializes on is not a fixed, guaranteed feature — the router reliably finds *some* real distinction to exploit, not always the *same* one.

Recommendation for the next run: `layers_to_replace=[0]` as the single best-evidenced candidate, with `[0, 4, 17, 21]` as a secondary option if testing more than one layer is worth the added complexity

— those three are the next tier by both average and consistency, despite 4 and 21 each collapsing on one of the two seeds tested. The five volatile mid-range layers (3, 5, 9, 14, 19) should be avoided in either case; their unpredictability makes them poor building blocks for a configuration meant to be reliable.

7. Current State and What Would Be Needed Before Scaling Up

What this phase established: the infrastructure-level failure modes are found and fixed — NaN divergence, checkpoint resume, disk exhaustion, and (the big one) the initialization bug that made every prior run's absolute numbers meaningless. One full run now completes cleanly end-to-end with sane losses throughout. Now that the missing control has been run (§6.4) *and* repeated with a second seed (§6.5): **at this scale and budget, the dense baseline beats the all-layers diffusion-MoE model decisively and consistently — better `task_loss`, an order-of-magnitude-plus better `val_ppl` on both seeds, and 15-21x less wall-clock time — while also being far more seed-consistent than the MoE configuration (sub-1% run-to-run variance vs. MoE's 44-52%).**

Remaining open questions:

1. The final near-zero `load_loss` coincides with the learning rate being fully decayed, on both seeds. It is plausible routing balance was actively, robustly resolving itself in both cases — and equally plausible it simply froze wherever it happened to be as gradients vanished toward the end of the cosine schedule, which would not be evidence of a stable equilibrium at all. Two data points now both showing the same end-of-run pattern makes the "real equilibrium" reading somewhat more credible than after §6.3 alone, but doesn't settle it.
2. 20M tokens is a smoke-test budget, not a real pretraining scale, and the standard case for why sparse MoE is worth its complexity (Clark et al. 2022's routed-vs-dense scaling laws, discussed earlier in this investigation) is specifically about quality-per-active-FLOP *at scale* — this comparison is nowhere near the regime that literature has actually demonstrated an advantage in. §6.4/§6.5's results do not show diffusion-MoE is a bad idea in general; they show this specific comparison offers no evidence for it, twice.

Recommendation, revised again: not ready for a scaled-up run, and the seed-repeat result (§6.5) removes the most likely objection to that conclusion — this was not one unlucky comparison. Before spending more compute on the current diffusion-MoE approach at all: (a) profile and reduce the CPU-bound Nyström/eigensolve cost specifically — a 15-21x wall-clock penalty needs a very large quality win to ever justify itself, and right now the result runs the other direction; (b) revisit whether all-layers replacement was the right configuration to keep pursuing — Phase 1's original oracle-ceiling work recommended a *selective* few layers, not all 24, which this investigation deliberately set aside to answer "what happens if every layer routes." That question has now been answered, fairly conclusively, at this scale: worse and less consistent than not routing at all. The selective-layer approach remains untested by this phase and is the more promising next direction, not a bigger version of what was just run.

8. Selective-Layer Results, the Refit-Discontinuity Investigation, and a Structural Alternative

Following §7's revised recommendation, this section covers what actually running §6.7's data-driven layer candidates showed, a real bug found while investigating why the routed layer wasn't recovering as expected, a deeper structural finding about *why* the recovery is incomplete even with that bug fixed, and the resulting decision to test a structurally different router as the next comparison.

8.1 Selective-layer results

Both of §6.7's candidates were run at the same 20M-token budget, `seed=42` , with the fixed initialization:

Config	task_loss	val_ppl	Wall-clock
Dense	4.80	93.2	20.1 min
Selective [0] (polar-leaf-25)	5.56	210.1	34.8 min
Selective [0,4,17,21] (smooth-donkey-26)	8.06	2,885.6	83.3 min
All-layers MoE	7.08	974.6	294.5 min

One well-chosen layer closes most of the gap to dense on every axis (`val_ppl` ~4.6x better than all-layers, wall-clock ~8.5x faster) — real support for the selective-layer direction over all-layers replacement. Adding the three second-tier candidates made things *worse*, not better — worse than [0] alone and worse than all-layers on `val_ppl` . This is a correction to §6.7's own framing: layers 4/17/21 were recommended based on *low seed-to-seed variance* in their `load_loss` , but their *average* `load_loss` was still moderate-to-poor (1.8–3.8, nowhere near layer 0's 0.21) — reliably mediocre is not the same as good, and diluting the one genuinely well-balanced layer with three consistently-so-so ones hurt more than it helped. Layer 0 alone remains the best MoE configuration found in this investigation.

8.2 Bug: `centroid_refresh_steps` was measured in micro-batches, not steps

Found by a sharp observation during live monitoring: a repeating loss-disruption pattern visible at 250-step intervals that didn't match the configured `centroid_refresh_steps=500` at all. Root cause:

`DiffusionMoELayer` 's internal `_step` counter increments once per `forward()` call, but `Trainer.train_step()` calls `forward()` once per *micro-batch* — `grad_accum_steps` times per logged/optimizer step, not once. Every other `*_steps` config value (`checkpoint_steps` , `eval_steps` , `log_steps`) is measured in optimizer-step units; `centroid_refresh_steps` silently wasn't. With this project's `grad_accum_steps=2` default, **every run in this investigation had been refitting landmarks twice as often as configured — 8 times over a 2,441-step run, not 4.**

Confirmed directly: the same crash-then-recover `load_loss` / `task_loss` signature already found at steps 500/1000/1500/2000 (§2.1) also appears identically at 250/750/1250/1750 in `graceful-universe-21` — exactly the halved-cadence prediction.

Fixed by threading `grad_accum_steps` into `DiffusionMoELayer` (and the full `_build_moe_layer` / `DiffusionMoETransformer` / `build_model_from_config` chain), so the refresh check is `_step % (centroid_refresh_steps * grad_accum_steps) == 0` — robust to future changes to `grad_accum_steps` , rather than requiring it to be manually doubled by hand.

8.3 The deeper finding: the model doesn't recover before the next refit hits

Investigating the disruption further (prompted by the question of whether a single refit's damage is transient or lasting) surfaced something worse than §2's original characterization. Tracing `task_loss` across a *full* refit-to-refit window in `graceful-universe-21` (step 250 to step 750):

```
step 230 (pre-refit): task_loss=6.92
step 270 (post-refit): task_loss=9.62    <- spike
step 490 (mid-window): task_loss=7.84    <- best it gets, still worse than pre-refit
step 750 (next refit): task_loss=8.45    <- never recovered, then hit again
```

`task_loss` never returns to its pre-refit level anywhere in that 500-step window, and the *next* refit lands before it can close the remaining gap. §2.1's original claim (" `load_loss` recovers within ~10-15 steps") was real but incomplete: that described `load_loss`'s aggregate *magnitude* returning to a normal noisy range, not the model's actual specialization quality recovering — the two were conflated. The honest framing: the diffusion map assumes something close to a stable manifold to fit a geometric embedding of, but the thing being embedded (post-attention activations) is itself continuously reshaped by the same gradient descent process the routing depends on (`task_loss`'s gradient flows into the attention weights that produce those activations; only the routing losses are detached — see `methodology.md` §5.2). A periodic *hard* refit repeatedly asks the router to re-orient to wherever that target has drifted to, with experts caught mid-specialization each time. This has a direct parallel in the deep-clustering literature, where alternating between fitting a clustering and updating the representation it's fit to is a known source of instability.

8.4 The sign-alignment fix helped only at the margin

A candidate mitigation: diffusion-map eigenvectors are only defined up to sign, and a refit replaces the basis outright (fresh landmarks, fresh eigensolve) with no guaranteed relationship to the old basis's orientation — `ExpertCentroids` doesn't get remapped when this happens, so an arbitrary sign flip adds a second, gratuitous discontinuity on top of the real drift from §8.3. Fixed in `NystromDiffusionMap.fit()` : snapshot the old basis's embedding of the current batch before overwriting anything, then flip each new component's sign to agree with it (confirmed against real data: a natural refit reproducibly flips a component's sign without the fix, dot product `-0.0043` against the pre-refit embedding, and does not with it).

Tested head-to-head on the best selective config (`layers_to_replace=[0]` , `seed=42`): `polar-leaf-25` (before) vs. `denim-darkness-27` (after).

	Before	After
<code>task_loss</code>	5.56	5.57
<code>val_ppl</code>	210.1	211.0
Wall-clock	34.8 min	34.8 min
Peak <code>task_loss</code> in the step-250 disruption window	7.60	7.15

Essentially no change to the final outcome, though a small, real softening of the immediate disruption (peak `task_loss` 7.60 → 7.15). This is exactly what §8.3's framing predicts: sign-alignment removes a secondary source of discontinuity, but the dominant cause — the representation itself moving under the router — is untouched by it. Kept (it's a correct, cheap, structural fix, and it does measurably soften the immediate jolt even if it doesn't close the gap), but not a solution to the underlying problem on its own.

8.5 The shared expert, made a uniform toggle

Originally added only to `DiffusionMoELayer` , unconditionally on (§3). Made a `use_shared_expert` config toggle (`routing.use_shared_expert` , default `true`) threaded uniformly across all four router variants (`diffusion` / `switch` / `cosine` / `random`), so it can be compared on/off consistently rather than assumed to help everywhere, and so the comparison in §8.6 holds it constant rather than present on only one side.

8.6 A structurally different router: Switch Transformer's learned linear gate

§8.3's finding reframes the problem: the issue isn't primarily the diffusion router's refit *mechanics* (§8.2/§8.4 both fixed real bugs there with only marginal effect) — it's that periodic hard refitting of a geometric embedding is fighting a representation that's continuously moving under gradient descent.

`SwitchMoELayer` — already implemented in this codebase as a baseline (`router: "switch"`), a single learned linear projection to expert logits, trained continuously by ordinary backprop, no diffusion map, no periodic refit, no discontinuity by construction — sidesteps this entire class of problem rather than mitigating it. Testing it at the same config as the best diffusion result (`layers_to_replace=[0]` , `use_shared_expert=true` , `seed=42`) directly asks whether the diffusion router's problems are specific to periodic hard refitting, or whether sparse routing itself is the harder part at this scale. *[Run in progress at time of writing; results to follow.]*

8.7 Observability gap: wandb never logged the actual run config

Found while verifying the Switch comparison (§8.6) was actually apples-to-apples with the diffusion runs on `top_k / n_experts` : `Trainer.__init__` called `wandb.init(project=...)` with no `config=` argument at all, for every run in this entire investigation. None of a run's actual hyperparameters — `routing.top_k` , `model.router` , `use_shared_expert` , any of it — were ever recorded in wandb. Confirming what a given run actually used required SSHing into the instance and reading its log file by hand (and even that was often unavailable in the moment: `nohup` 's stdout redirection is block-buffered, so the printed config frequently hadn't flushed to disk yet for a still-running process — the same buffering behavior noted throughout this investigation's live-monitoring sections).

Fixed: `wandb.init(..., config=wandb_config)` , where `wandb_config` is `OmegaConf.to_container(config, resolve=True)` for a real Hydra `DictConfig` (flattened into a plain, JSON-serializable dict — passing the `DictConfig` object through as-is would not have logged correctly), falling back to the config object as-is for the plain-dict configs the test suite uses. Every run from this point forward has its actual config in wandb's own config panel — no more reconstructing it from launch commands or SSH sessions after the fact.

9. Code Changes Summary

File	Change
<code>geometry/nystrom.py</code>	Guarded the <code>d_alpha_new</code> division against exact-zero underflow; added live <code>eps_/landmark-spectrum</code> refresh in <code>transform()</code> (new <code>_refresh_eps</code> param, default on); extracted <code>_fit_landmark_spectrum</code> shared between <code>fit()</code> and the refresh path; <code>fit()</code> now sign-aligns the freshly-fit eigenbasis to agree with the previous one's embedding of the same batch (§8.4)
<code>geometry/eigensolver.py</code>	<code>diffusion_eigenvectors</code> gained a <code>random_state</code> param, seeding ARPACK's starting vector — needed once the eigensolve could rerun mid-training, not just once per landmark set
<code>models/moe_layer.py</code>	Added <code>shared_expert</code> (outside the router; §3); <code>_compute_diffusion_coords</code> now also refits when <code>ndm.landmarks_</code> is <code>None</code> (checkpoint-resume fix), no only on the step-count schedule; refresh cadence now accounts for <code>grad_accum_steps</code> (§8.2); <code>use_shared_expert/grad_accum_steps</code> constructor params

models/switch_moe_layer.py, models/cosine_moe_layer.py, models/random_moe_layer.py	Added the same use_shared_expert toggle (§8.5) — the shared expert is no longer DiffusionMoELayer-only
models/moe_model.py	Fixed token_embedding init (std=0.02, was nn.Embedding default std=1.0); added depth-aware $1/\sqrt{2 \times n_layers}$ scaling to every out_proj/down_proj weight after construction; threads grad_accum_steps/use_shared_expert through _build_moe_layer
models/model_factory.py	Reads training.grad_accum_steps and routing.use_shared_expert from config
training/trainer.py	Added non-finite-loss guard in train_step (raises FloatingPointError before backward()/optimizer.step() instead of silently training on NaN/Inf added _prune_old_checkpoints (keep_last_n_checkpoints, default 2); wandb.init now actually passes config= (§8.7) — was never logging any run's real hyperparameters at all
data/dataloader.py	Fixed local_data_files reaching the real Hydra-config-driven training path — a ConfigAttributeError several frames deep (omegaconf.ListConfig vs. plain list), not an obvious error at the call site
scripts/expert_attribution.py	Added Hydra-style CLI override support (parse_known_args), so it can point at the real data config a checkpoint was trained against instead of always falling back to base_config.yaml's defaults
scripts/train.py	cfg.seed now actually applied (torch.manual_seed/np.random.seed/random.seed/torch.cuda.manual_seed_all before model construction — previously a no-op
configs/base_config.yaml	routing.mu_load: 0.01 → 2.0 (measured, §6.2); training.keep_last_n_checkpoints: 2 (new); routing.use_shared_expert: true (new, §8.5)

All changes are covered by tests reproducing the specific failure before confirming the fix, not just testing the fix in isolation — e.g. a mixed-batch outlier reproducing the exact NaN-underflow condition; a cross-object checkpoint resume at a step deliberately off the refit schedule, confirmed to crash on the pre-fix code (verified by temporarily reverting the fix) and pass after; a monkeypatched zero-contribution routed path confirming the shared expert's contribution is structurally unconditional; a real disk-bounded checkpoint-pruning scenario; the random-label cross-entropy check confirming init-time loss lands near the true random baseline (confirmed to fail — 61.5 vs. an expected <13.8 — on the pre-fix code); a same-seed-reproduces/different-seed-diverges check on model initialization; a centroid_refresh_steps / grad_accum_steps interaction check confirming the correct (not halved) refit cadence; and a sign-alignment check confirming a real refit reproducibly flips a component's sign without the fix and does not with it. Full suite: 385 tests passing at last check.

Appendix: Open Questions Carried Forward

- Does the depth-progression / sequence-start routing pattern (§6.6) persist, sharpen into content-sensitivity, or dissolve entirely at a later training step and a larger token budget? **Partially answered for layer 0 by §6.7:** by the final checkpoint it had moved past pure position into a real content-correlated split (digit vs. word tokens on one seed), though the specific feature wasn't

identical on the other seed. Layer 6 specifically (the depth-progression layer) wasn't re-checked at the final checkpoint — still open.

- Should `nu_sep` (currently unchanged at 0.05) be measured the same way `mu_load` was (§6.2)? `sep_loss` showed real instability of its own (swings from -3.5 to -10.6 within ~120 steps in the collapsed run) that was not directly investigated — it may be a downstream symptom of the same collapse dynamic, or a separate miscalibration.
- The self-tuning (per-point adaptive) bandwidth design discussed but not implemented (the "Option 2" alternative to §2.2's global live-refresh) remains a candidate if staleness-driven instability recurs even with the current fix.
- `NystromDiffusionMap`'s own k-means/eigensolve randomness still has a hardcoded `random_state=42`, independent of `cfg.seed` (§9) — the seed-repeat check in §6.5 varied model init and data-adjacent RNG state, but not this. Worth wiring through too if the seed-sensitivity finding in §6.5 needs isolating further (is it driven by init, by the diffusion-map's own landmark selection, or both).